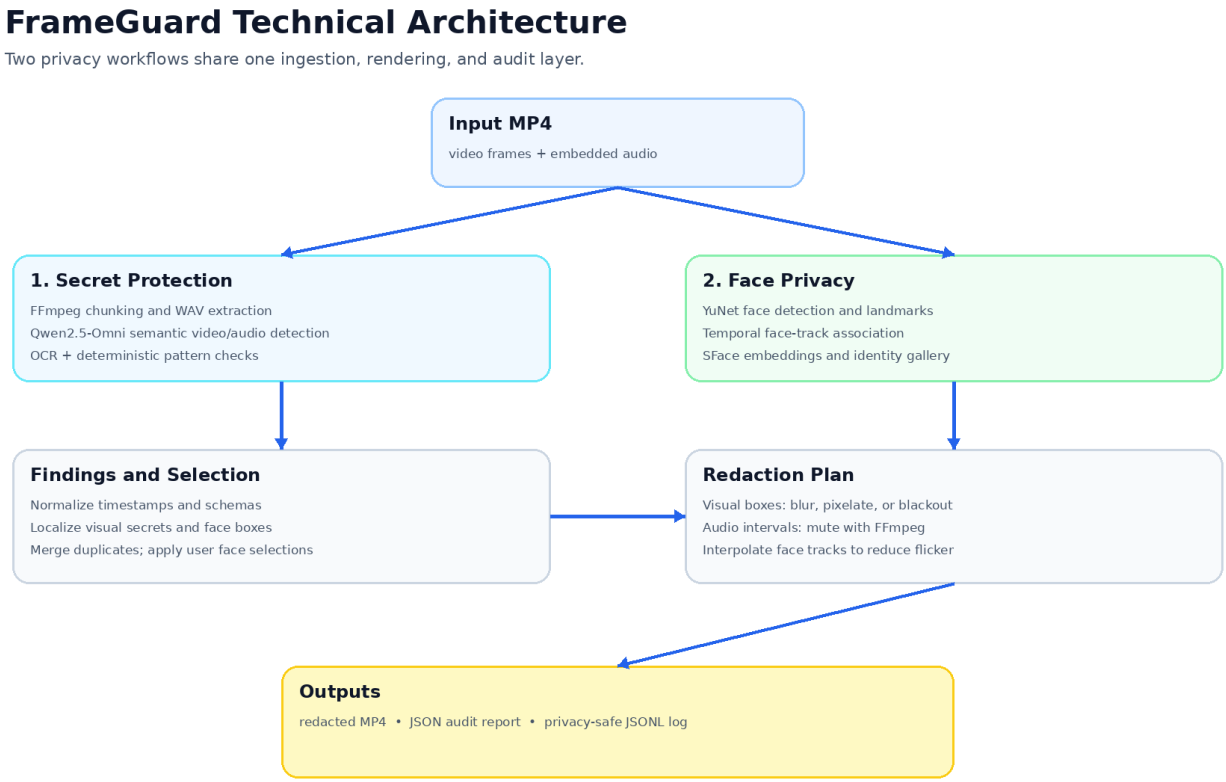
FrameGuard

Multimodal video privacy for sensitive content and human faces

MULTIMODAL AI SUBMISSION



Two independent privacy workflows

Secret Protection detects visible and spoken sensitive information. Face Privacy detects people, groups repeated face tracks, and blurs user-selected identities.

1. Project overview

FrameGuard analyzes video and audio, locates privacy-sensitive content, and produces a redacted MP4 with a machine-readable audit trail. The product separates two distinct user goals into dedicated workflows.

Workflow	Primary purpose	Result
Secret Protection	Detect and redact credentials, personal data, internal identifiers, QR codes, and spoken sensitive information.	Localized visual blur and muted audio intervals.
Face Privacy	Detect people, deduplicate face-track fragments, and apply user-controlled blur rules.	Stable face redaction for selected identities.

Shared output layer

- Redacted MP4 with non-sensitive content preserved.
- JSON audit report containing timing, localization, confidence, action, and detector sources.
- Privacy-safe JSONL run log containing metrics and execution events without raw secret values or face embeddings.

Interface structure

```
Sensitive Content  -> Detect -> Review -> Redact
Face Privacy       -> Detect People -> Select Profiles -> Redact
Results            -> Preview -> Download -> Audit
Settings           -> Models -> Thresholds -> Logging
```

2. Secret Protection

Secret Protection inspects visible frames and spoken audio. It combines semantic multimodal analysis with deterministic OCR and structured-pattern validation so that contextual and exact-value detection can operate together.



Visible credentials are localized and blurred while surrounding screen content remains readable.

Detection pipeline

- FFmpeg splits the MP4 into short analysis chunks and extracts mono WAV audio.
- Qwen2.5-Omni-3B receives video and audio through an OpenAI-compatible vLLM endpoint.
- The model returns structured findings containing type, modality, timing, confidence, and reason.
- OCR extracts exact text and word-level bounding boxes from sampled frames.
- Deterministic patterns validate emails, IP addresses, credentials, phone numbers, identifiers, and private URLs.
- QR detection supplies exact image coordinates for encoded content.
- Findings from all detectors are normalized, merged, and deduplicated.

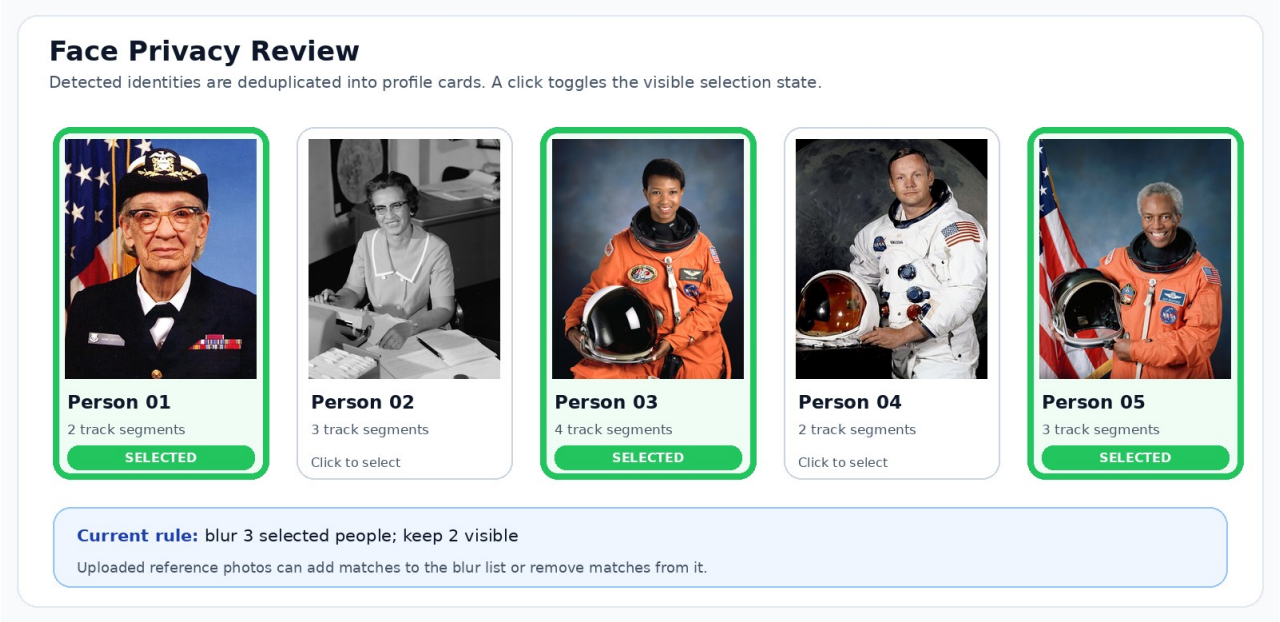
Redaction behavior

Visual findings are rendered as Gaussian blur, pixelation, or blackout over localized regions. Spoken findings are converted into padded audio intervals and muted with FFmpeg. The visual and audio streams are then assembled into the final MP4.

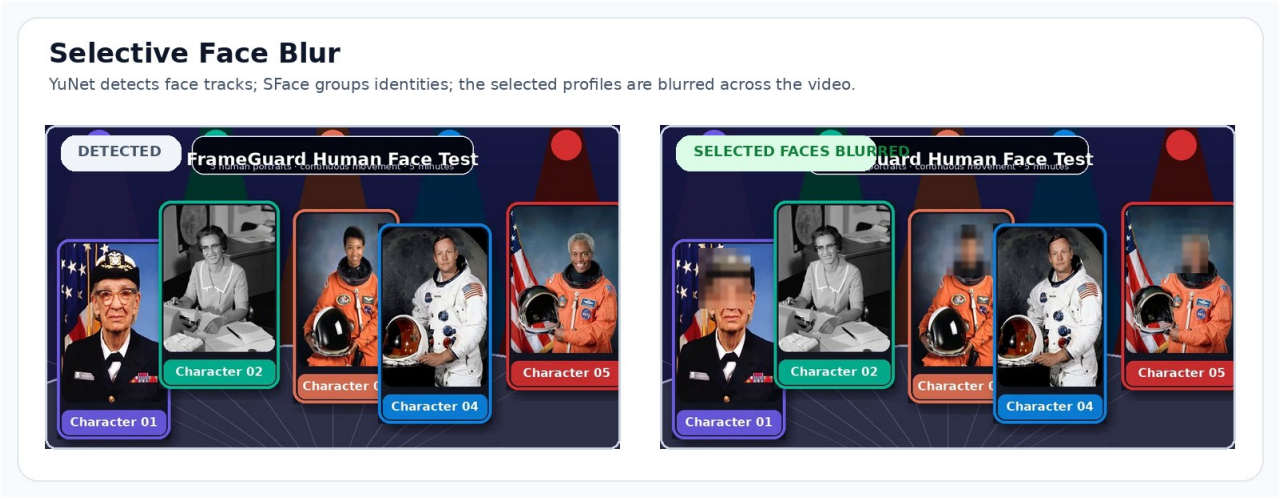
Hybrid responsibility split
Qwen determines what is sensitive and why. OCR, QR detection, and deterministic patterns determine where the sensitive content appears and verify common structured formats.

3. Face Privacy

Face Privacy provides deterministic user control over which people are blurred. It does not require automatic age classification or identity naming. The user reviews anonymous profile cards and applies explicit blur or keep-visible rules.



Deduplicated face profiles with a visible selected state and a clear blur rule.



Selected profiles are mapped back to their temporal face tracks and blurred throughout the video.

Face detection and tracking

- YuNet produces face boxes, confidence scores, and five facial landmarks on sampled frames.
- Temporal association links detections using geometry, overlap, motion continuity, and maximum-gap rules.
- Short unstable tracks are rejected; accepted tracks retain timestamped box observations.
- The renderer interpolates box positions between sampled observations to reduce flicker.

Identity gallery and reference photos

- Several candidate frames are ranked using source face size, sharpness, and YuNet confidence.
- SFace aligns the strongest crops and creates normalized embeddings.
- Multiple strong embeddings are averaged for each track fragment.
- Compatible, non-overlapping fragments are merged into anonymous identity profiles.
- Uploaded photos are encoded through the same YuNet and SFace path and matched by cosine similarity.
- Matches remain user-reviewed and are not treated as identity verification.

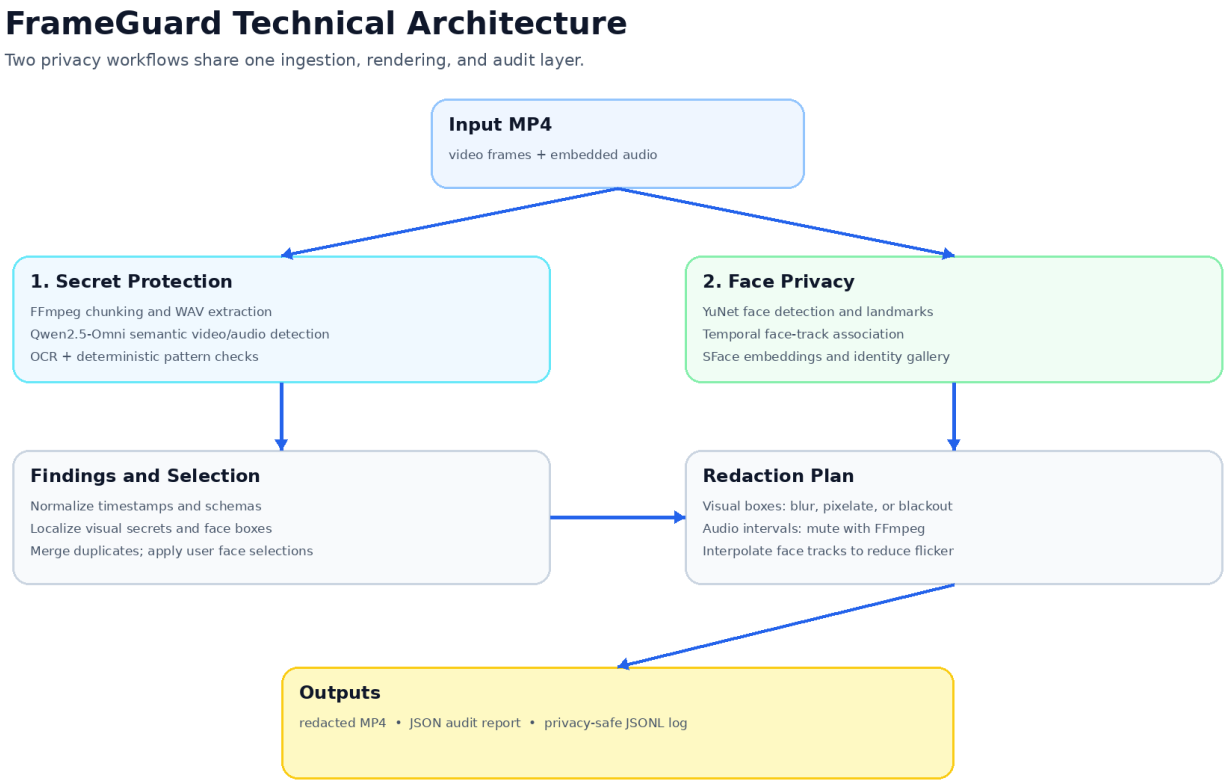
Selection modes

Mode	Behavior
Blur selected	Only checked gallery profiles and accepted uploaded-photo matches are blurred.
Keep selected visible	All detected profiles are blurred except the checked profiles and accepted keep-visible matches.

User control over inference

The gallery is the final authority. Automatic grouping and uploaded-photo similarity narrow the review task, but the user decides which identities are blurred or kept visible.

4. Technical architecture



Common finding schema

All detectors produce the same normalized representation, allowing secrets and faces to share the renderer and audit system.

```
{
  "type": "api_key | email | face | qr_code | ...",
  "modality": "visual | audio | both",
  "start_ms": 0,
  "end_ms": 5000,
  "confidence": 0.97,
  "action": "blur | pixelate | blackout | mute",
  "sources": ["qwen", "ocr", "regex", "yunet", "sface"],
  "observations": [{"time_ms": 0, "x": 0, "y": 0, "width": 100, "height": 40}]
}
```

Core modules

Module	Role	Module	Role
app.py	UI and workflow state	pipeline.py	Analysis orchestration
multimodal_llm.py	Qwen/vLLM requests	visual_locator.py	OCR localization
face_tracking.py	YuNet face tracks	face_reference.py	SFace matching
face_gallery.py	Profile review policy	redact.py	Visual/audio rendering
observability.py	Logs and metrics	schemas.py	Shared data models

5. Audit, privacy, and outputs



The audit layer records redaction decisions without persisting raw secret values or face embeddings by default.

Output files

```
outputs/
|-- <video>_redacted_<run-id>.mp4
|-- <video>_report_<run-id>.json
`-- logs/
    `-- <video>_<run-id>.jsonl
```

Privacy controls

- Exact secret values are suppressed from normal logs.
- Raw multimodal model output is excluded by default.
- Face crops and SFace embeddings remain in the active application session.
- Uploaded reference photos are not embedded in the audit report.
- Reports use anonymous profile IDs and track IDs rather than names.
- The audit report records actions, timing, localization, detector sources, and match scores.

Evaluation

Secret Protection	Face Privacy
Secret recall by type	Face-detection recall
Visual localization accuracy	Track stability
Audio interval accuracy	Duplicate-profile rate
False-redaction rate	Reference-match accuracy
Rendered-output verification	Blur continuity and flicker

Known limitations

- Very small, blurred, profile, or occluded faces may not produce clear gallery images.
- Fast motion and long occlusion can fragment face tracks.
- SFace supports user-assisted grouping and matching; it is not identity verification.
- OCR depends on resolution, contrast, font size, and motion blur.
- Semantic detection may miss unusual secret formats; deterministic patterns provide partial coverage.
- The rendered video must be reviewed because detector metrics alone do not prove successful redaction.

6. Submission summary

FrameGuard is a multimodal privacy system built around two clear product paths. Secret Protection combines video, audio, OCR, pattern recognition, localization, and media transformation. Face Privacy combines neural face detection, temporal tracking, face embeddings, user-reviewed identity selection, and track-aware rendering.

Primary contribution
A single auditable redaction engine accepts findings from semantic multimodal analysis, deterministic text detectors, QR localization, face tracking, and user-reviewed identity selection.

Run

```
./scripts/start.sh
```

Submission repository policy

The root README is the authoritative project document. Temporary patch instructions, environment-specific guides, phase notes, and redundant nested READMEs are excluded from the submission repository.